

Stefano Roy Bisignano

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AI Engineer specializing in **LLMs**, **ML systems**, and robotics-oriented generative AI. I build fine-tuning, evaluation, and multimodal pipelines with **LLMOps** for production-ready models. I care about clear metrics and real-world impact.

Personal: I like building end-to-end systems and writing concise technical notes. Outside work I train regularly and spend weekends skiing when possible.

EXPERIENCE

ML Engineer → ML Engineer Lead (promoted)

May 2024 – Oct 2025

Simplex Rapid

Milan, Italy

- Integrated LLM and AI systems into precision mechanics workflows covering model selection, domain adaptation, deployment, observability, and cost/latency optimization.
- Built a multilingual translation pipeline with fine-tuned models across **20 languages**; achieved **~53% touchless rate** and reduced human effort from **~8,000h** to **~2,350h** (**~71% reduction**). Estimated **~76% cost reduction** based on QA sampling and project logs.
- Migrated an agentic business assistant to the Responses API with tool calling, monitoring, and tracing for reliability and auditability.
- Combined LLM and vision to extract technical parameters from spring drawings/PDFs with per-field validation and human-in-the-loop review, enabling automatic population of manufacturing data.

PROJECTS

Edge-VLA-Micro. Built a voice-controlled FPV drone agent: camera + speech input feed a VLM action proposal, symbolic HSV/Pydantic guards block hallucinated moves, and validated commands execute through MAVSDK/PX4; optimized TTFT by 65.6%. **2026**

Humanoid-Motion-Diffusion. Audio-conditioned DDPM Transformer for SMPL/AIST++ humanoid motion; biomechanical evaluation reduced TSI 12.60 to 0.08. **2026**

SSL for RCC Classification. Self-supervised learning on histopathology WSIs for subtype discovery; preliminary phase completed with **30/30 with honors**; paper in progress; pending publication. **2026**

Affordance 3D Highlighting. Zero-shot localization of functional 3D regions from language prompts, inspired by CLIP and 3D Highlighter (CVPR 2023). **2025**

Embedded Vision Trade-offs (Arduino Nicla Vision). Hardware-in-the-loop benchmark suite exposing deployment trade-offs beyond validation accuracy; protocol-driven evaluation and reproducible runs. **2026**

Swiss AI Weeks (ETH AI Center 2025). Multi-agent dialog-to-action system with disagreement arbitration and safe human handoff for financial advisory tasks. **2025**

OnlyFly (HackUPC 2025, 4th/150+). AI home-exchange MVP integrating preference matching, flight search (Skyscanner), and payments (Revolut). **2025**

PostGenius (GenAI.Works 2024, 7th/4,500). RAG-driven multimedia generator (text, image, video) built with OpenAI, Vectara, and Runway APIs. **2024**

EDUCATION

MEng in Artificial Intelligence and Data Analytics, Politecnico di Torino, Italy

Apr 2024 – Dec 2026

BSc in Computer Engineering, Politecnico di Torino, Italy

Oct 2020 – Apr 2024

SKILLS & INTERESTS

AI Systems & Opt:	Inference Optimization, Model Quantization, LLM Architectures, Multi-Agent Systems
Programming:	Python, C/C++, Rust, CUDA, Triton, Bash, PyTorch, JAX
Robotics & Sim:	ROS2, MuJoCo, Gymnasium, Gazebo, Physics-Informed Simulations
Infra & MLOps:	Docker, Kubernetes, CI/CD Pipelines, FastAPI, Triton Inference Server, Production Monitoring
Languages:	Italian (Native), English (Fluent), German (Basic), Spanish (Basic)
Interests:	Entrepreneurship, Politics, History, Physics, Neuroscience, Robotics, Freestyle Skiing